

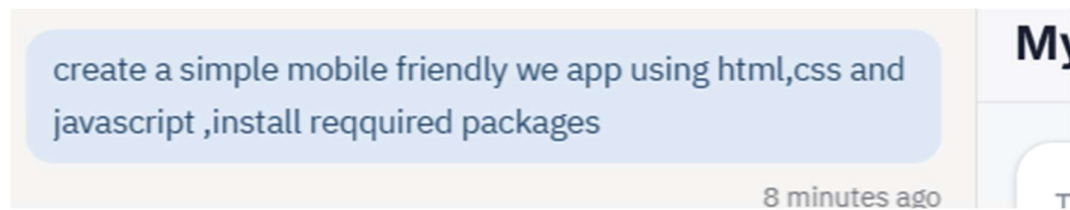
ASS-24.1(AI-AC)

2303A51804

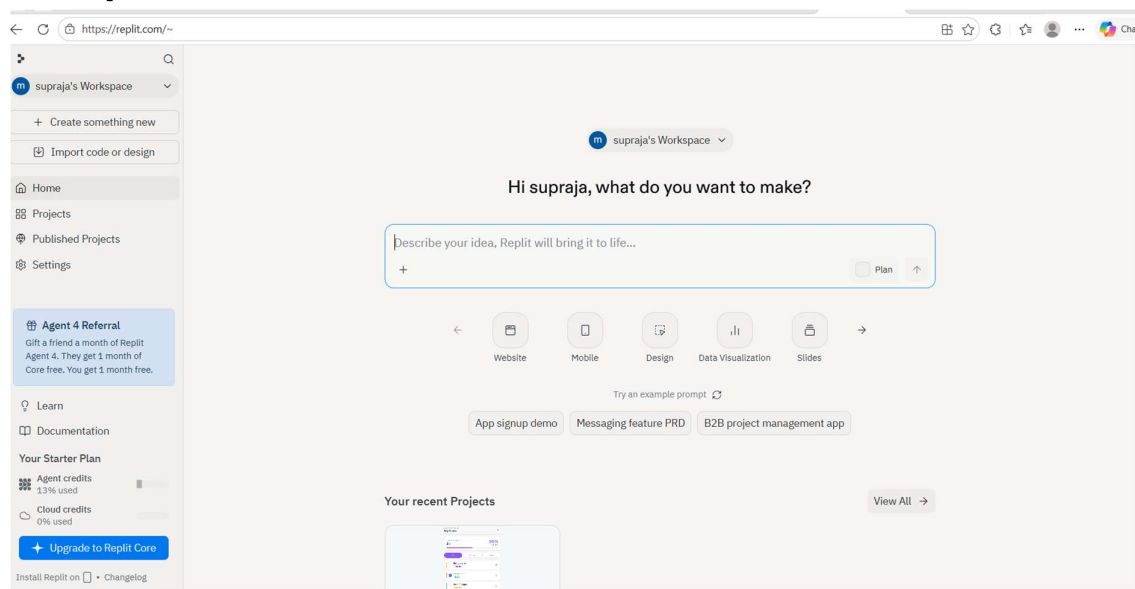
Batch-28

Task-1:Environment Setup

Prompt:



Setup:



Explanation:

1. A free account was created in Replit to use it as a cloud-based IDE.
2. A new project (Repl) was created by selecting the HTML, CSS, and JavaScript template.
3. Different templates like React Native, Flutter, and HTML/CSS/JS were explored for mobile app development.

4. The Replit interface, including editor, console, and file structure, was understood.
5. The environment was prepared for building and testing mobile applications.

Task-2: Basic Mobile App

Prompt:

display app title "Hello App" input field to enter user's name, Button to submit, show greeting message "Hello,[name]" dynamically and keep it simple and responsive for mobile

Code:

```
lib > api-client-react > package.json
1  {
2    "name": "@workspace/api-client-react",
3    "version": "0.0.0",
4    "private": true,
5    "type": "module",
6    "exports": {
7      ".": "./src/index.ts"
8    },
9    "dependencies": {
10     "@tanstack/react-query": "catalog:"
11   },
12   "peerDependencies": {
13     "react": ">=18"
14   }
15 }
16
```

lib > api-spec > openapi.yaml

```
1 openapi: 3.1.0
2 info:
3   # Do not change the title, if the title changes, the import paths will be broken
4   title: Api
5   version: 0.1.0
6   description: API specification
7 servers:
8   - url: /api
9     description: Base API path
10 tags:
11   - name: health
12     description: Health operations
13 paths:
14   /healthz:
15     get:
16       operationId: healthCheck
17       tags: [health]
18       summary: Health check
19       description: Returns server health status
20       responses:
21         "200":
22           description: Healthy
23           content:
24             application/json:
25               schema:
26                 $ref: "#/components/schemas/HealthStatus"
27 components:
28   schemas:
29     HealthStatus:
30       type: object
31       properties:
```

```
import { defineConfig, InputTransformerFn } from "orval";
import path from "path";

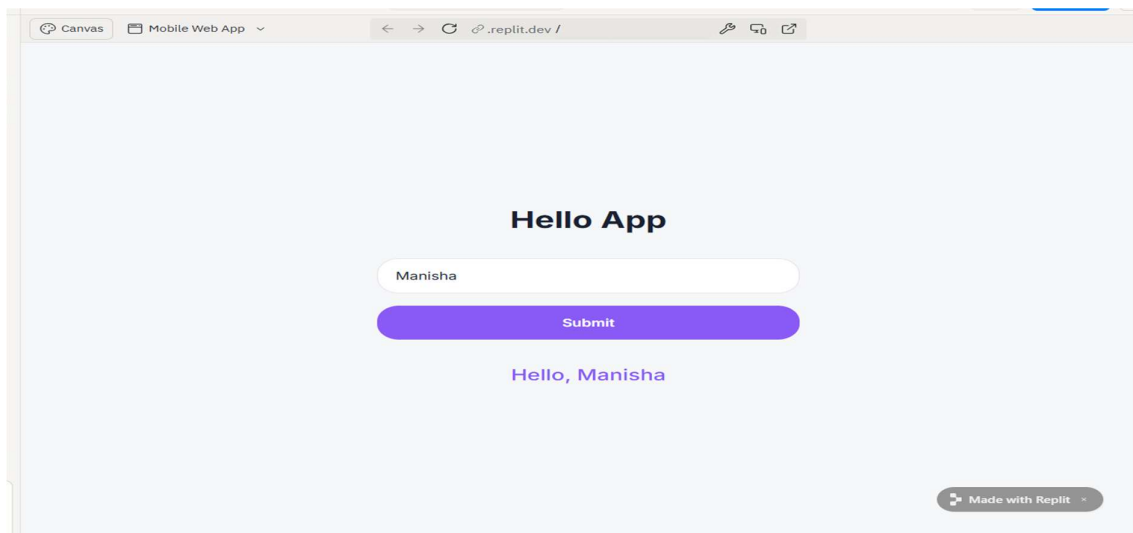
const root = path.resolve(__dirname, "..", "..");
const apiClientReactSrc = path.resolve(root, "lib", "api-client-react", "src");
const apiZodSrc = path.resolve(root, "lib", "api-zod", "src");

// Our exports make assumptions about the title of the API being "Api" (i.e. generated)
const titleTransformer: InputTransformerFn = (config) => {
  config.info ??= {};
  config.info.title = "Api";

  return config;
};

export default defineConfig({
  "api-client-react": {
    input: {
      target: "./openapi.yaml",
      override: {
        transformer: titleTransformer,
      },
    },
    output: {
      workspace: apiClientReactSrc,
      target: "generated",
      client: "react-query",
      mode: "split",
      baseUrl: "/api",
      clean: true,
    },
  },
});
```

Output:



1. Explanation: A simple mobile-friendly application called “Hello App” was created using HTML, CSS, and JavaScript in Replit.
2. The app includes an input field where the user can enter their name.
3. A button is provided to submit the input entered by the user.
4. When the button is clicked, a greeting message like “Hello, [name]” is displayed dynamically on the screen.
5. AI assistance in Replit was used to generate the basic structure (boilerplate code) and to understand how the app works

Task-3: Adding Features with AI Assistance

Prompt:

Enhance this mobile app by adding: Navigation bar(Home,about,contact) and a simple user registration form ,Basic styling for mobile screens and add comments(documentation) for each function

Code:

```
1  import { useState } from "react";
2
3  /**
4   * Home page component.
5   * Displays a welcome greeting and a name input with dynamic "Hello, [name]" resp
6   */
7  export default function Home() {
8    /** The current value of the name input field. */
9    const [name, setName] = useState("");
10
11    /** The greeting message shown after the user submits their name. */
12    const [greeting, setGreeting] = useState("");
13
14    /**
15     * Handles form submission.
16     * Validates the name field and sets the greeting message.
17     * @param e - The form submit event.
18     */
19    function handleSubmit(e: React.FormEvent) {
20      e.preventDefault();
21      if (name.trim()) {
22        setGreeting(`Hello, ${name.trim()} 🍌`);
23      }
24    }
25
26    /**
27     * Clears the greeting and resets the input field.
28     */
29    function handleReset() {
30      setName("");
```

artifacts > mobile-app > src > pages > Contact.tsx > ...

```
1  import { useState } from "react";
2
3  /** Represents the shape of the registration form data. */
4  type FormData = {
5    fullName: string;
6    email: string;
7    phone: string;
8    password: string;
9  };
10
11 /** Represents form field validation errors. */
12 type FormErrors = Partial<Record<keyof FormData, string>>;
13
14 /**
15  * Contact page component.
16  * Contains a user registration form with client-side validation.
17  * On successful submission, shows a confirmation message.
18  */
19 export default function Contact() {
20   /** Form field values managed as a single state object. */
21   const [form, setForm] = useState<FormData>({
22     fullName: "",
23     email: "",
24     phone: "",
25     password: "",
26   });
27
28   /** Validation error messages per field. */
29   const [errors, setErrors] = useState<FormErrors>({});
30
```

```

artifacts > mobile-app > src > pages > About.tsx > ...
1  /**
2   * About page component.
3   * Displays a brief description of the Hello App and its features.
4   */
5  export default function About() {
6    return (
7      <main className="flex flex-col items-center min-h-[calc(100vh-56px)] px-6 py-10 bg-background">
8        <div className="w-full max-w-sm">
9
10         {/* Page heading */}
11         <h2 className="text-2xl font-bold text-foreground mb-1">About</h2>
12         <p className="text-sm text-muted-foreground mb-8">Learn more about this app.</p>
13
14         {/* App description card */}
15         <div className="bg-card border border-border rounded-2xl p-5 mb-4 shadow-sm">
16           <h3 className="text-base font-semibold text-foreground mb-2">What is Hello App?</h3>
17           <p className="text-sm text-muted-foreground leading-relaxed">
18             Hello App is a simple, mobile-friendly web application that demonstrates
19             modern React development with clean UI patterns and smooth interactions.
20           </p>
21         </div>
22
23         {/* Features list card */}
24         <div className="bg-card border border-border rounded-2xl p-5 mb-4 shadow-sm">
25           <h3 className="text-base font-semibold text-foreground mb-3">Features</h3>
26           <ul className="flex flex-col gap-2">
27             {[
28               "Personalized greeting on the Home page",
29               "User registration form with validation",
30               "Responsive mobile-first design",

```

Output:

Hello App
[Home](#)
[About](#)
[Contact](#)

Welcome

Enter your name to get a personalized greeting.

Say Hello

Made with Replit

Explanation:

1. The initial project was created using a Node.js/TypeScript template instead of HTML/CSS/JavaScript.
2. Because of this, files like index.html, style.css, and script.js were not available.
3. The project structure showed backend files such as package.json and .ts files, which are not suitable for this task.
4. To fix this, a new Replit project was created using the HTML, CSS, and JavaScript template.
5. This allowed proper access to front-end code and made it easier to build the mobile web application.

Task-4: Testing on Mobile

Testing:

The screenshot shows a mobile web application interface. At the top, there is a navigation bar with the text 'Hello App' on the left and three links: 'Home' (underlined), 'About', and 'Contact'. Below the navigation bar is a large light blue area. In the center of this area, the word 'Welcome' is displayed in bold. Below it, a smaller text says 'Enter your name to get a personalized greeting.' There is a white input field with the text 'Manisha' inside. Below the input field is a purple button with the text 'Say Hello'. Below the button is a light purple box containing the text 'Hello, Manisha' followed by a yellow smiley face emoji. Below this box is a small link that says 'Reset'. In the bottom right corner of the light blue area, there is a small dark grey button with the text 'Made with Replit' and a small 'x' icon.

About

Learn more about this app.

What is Hello App?

Hello App is a simple, mobile-friendly web application that demonstrates modern React development with clean UI patterns and smooth interactions.

Features

- Personalized greeting on the Home page
- User registration form with validation
- Responsive mobile-first design
- Clean navigation between pages

Version 1.0.0 — Built with React + Vite

Made with Replit



Registration Successful!

Welcome, **Boddu Manisha**! Your account has been created.

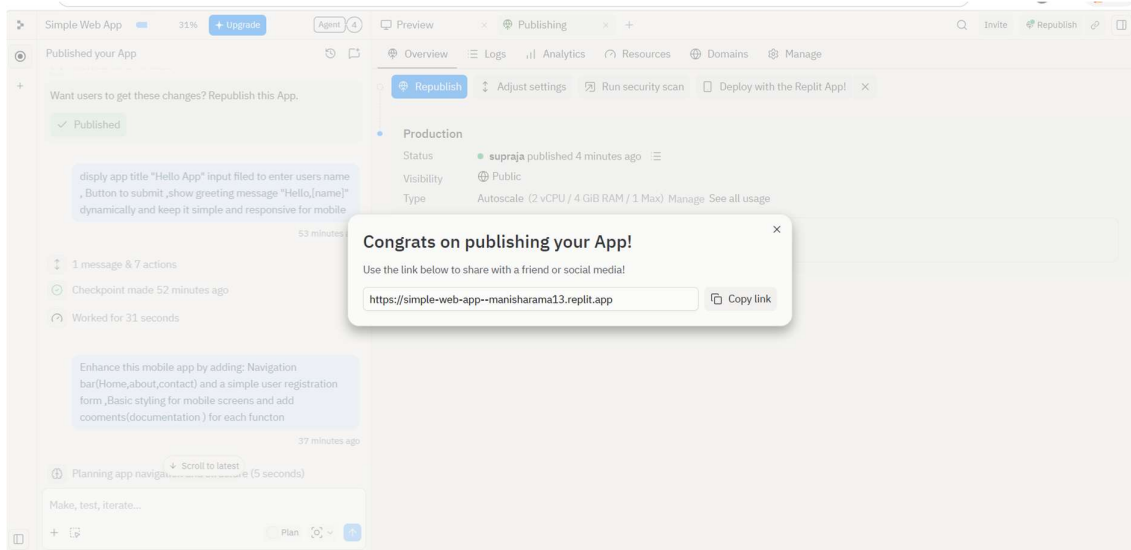
Register Another

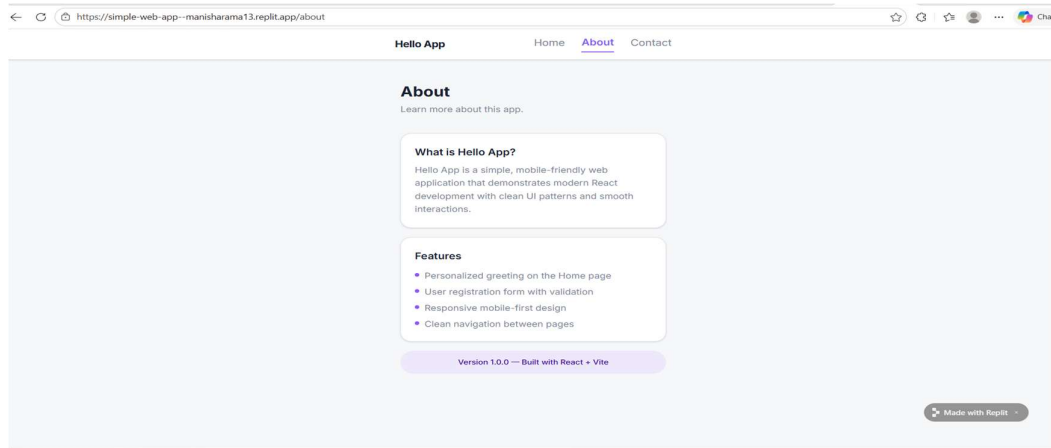
Made with Replit

Explanation:

1. The application was run in Replit using the mobile preview mode.
2. The app was tested on different screen sizes to ensure proper responsiveness.
3. All UI elements like buttons, text, and navigation were checked for alignment.
4. Minor UI issues such as spacing and layout were identified and improved.
5. AI suggestions were used to fix alignment and enhance the mobile user experience.

Task-5: Deployment





Explanation:

1. The application was deployed using Replit's hosting feature.
2. The "Publish Now" option was used to generate a public URL for the app.
3. The generated link was opened in a browser to test the application.
4. The app was successfully accessible online from any device.
5. The deployment link was shared with others for testing and feedback.